5.1.2

(d) $\overline{x\overline{y} + y\overline{z} + \overline{x}z} = (\overline{x\overline{y}})(\overline{y\overline{z}})(\overline{\overline{x}z}) = (\overline{x} + y)(\overline{y} + z)(x + \overline{z})$

5.3.2

(a) DNF

(b) Both

(c) CNF

(d) Neither # 5.3.3

(e) $(x + y\overline{z})(\overline{x} + y + \overline{z}) = x\overline{x} + xy + x\overline{z} + \overline{x}y\overline{z} + y\overline{z} + y\overline{z}$ (distributive) = $xy + x\overline{z} + \overline{x}y\overline{z} + y\overline{z}$ (complement, idempotent) = $xy + x\overline{z} + y\overline{z}$ (distributive, annulment)

5.4.1

(b) $f = \overline{x\overline{y}}z + xy\overline{z}$ (minterms $m_1 + m_6$) Only multiplication and complement: $f = \overline{\overline{x\overline{y}}z \cdot \overline{xy\overline{z}}}$ Only addition and complement: $f = \overline{x + y + \overline{z}} + \overline{\overline{x} + \overline{y} + z}$

5.4.2

(a) $f = x\overline{y} + \overline{x}y$ (XOR, minterms $m_1 + m_2$) Only NAND: $f = \overline{\overline{x \cdot \overline{x\overline{y}}} \cdot \overline{y \cdot \overline{xy}}}$ Only NOR: $f = \overline{\overline{x + y} + \overline{\overline{x + x} + \overline{y + y}}}$

5.5.1

(a) **Satisfied** when $x = True$, $y = False$, and $z = False$

(b) **Satisfied** when $x = True$, $y = False$, and $z = False$

(c) **Not satisfiable.**

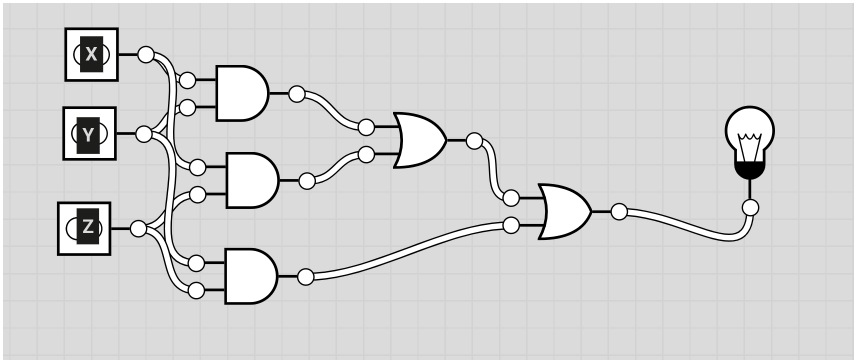
x	y	z	$f(x, y, z)$
False	False	False	False
False	False	True	False

x	y	z	$f(x, y, z)$
False	True	False	False
False	True	True	False
True	False	False	False
True	False	True	False
True	True	False	False
True	True	True	False

(d) **Satisfied** when $x = True$, $y = False$, and $z = False$

5.6.4

(b) $f = xy+xz+yz$



5.4.5

(a) We will construct {+, complement} which is functional complete. Since the complement operation is in both sets, it is sufficient to show that + can be computed using complement and $\rightarrow$. One of the conditional identities from the laws of propositional logic says that $x \rightarrow y$ is equivalent to $\overline{x} + y$. Therefore, $x + y$ is equivalent to $\overline{x} \rightarrow y$. Therefore, {complement, $\rightarrow$} is functional complete. # 5.5.5

(b) $x_{A1} + x_{A2} + x_{A3}$ (1 iff A is in at least one period)

(c) $\overline{x_{A1}x_{A2} + x_{A1}x_{A3} + x_{A2}x_{A3}}$ (1 iff A is in at most one period)

(d) $\overline{x_{A1}x_{B1}} \cdot \overline{x_{A2}x_{B2}} \cdot \overline{x_{A3}x_{B3}}$ (1 iff A and B not in same period)

(e) $4n + 3m$ terms: n at-least-one clauses (1 per class) + $3n$ at-most-one clauses (3 pairwise clauses per class) + $3m$ incompatibility clauses (3 periods per forbidden pair)

5.6.3

(d) List the the law you apply. Please don't forget to draw the simplified circuit

$F(x,y) = \overline{\overline{x} \cdot \overline{y}} + \overline{y + \overline{y}} = \overline{\overline{x} \cdot \overline{y}} + \overline{1}$ (Complement Law) $= \overline{\overline{x} \cdot \overline{y}} + 0$ (Substitution) $= (\overline{\overline{x}} + \overline{\overline{y}}) + 0$ (De Morgan's Law) $= (x + y) + 0$ (Double Negation Law) $= x + y$ (Identity Law)

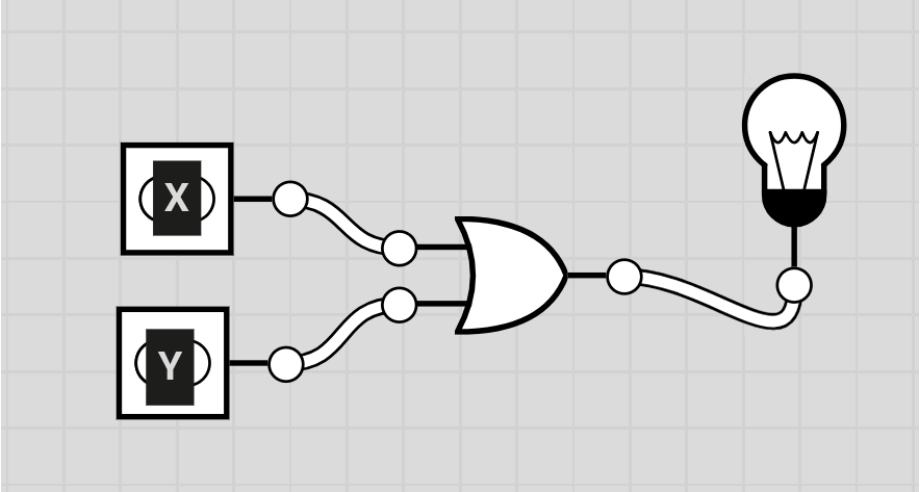


Figure 1: Pasted image 20261101171947.png